

TOP 50 IMPORTANT ELECTRICAL FORMULAS

Basic Formula

1. $V = I \times R$
2. $I = V / R$
3. $R = V / I$

Power Formulas

4. $P = V \times I$
5. $P = I^2 \times R$
6. $P = V^2 / R$

Energy

7. $E = P \times t$

Electric Charge

8. $Q = I \times t$

Current

9. $I = Q / t$

Resistance of Conductor

10. $R = \rho L / A$

Joule's Law

11. $H = I^2 \times R \times t$

Angular Frequency

12. $\omega = 2\pi f$

INDUCTIVE REACTANCE

14. $X_L = 2\pi fL$

CAPACITIVE REACTANCE

15. $X_C = 1 / (2\pi fC)$

IMPEDANCE

16. $Z = \sqrt{R^2 + (X_L - X_C)^2}$

POWER FACTOR

17. $PF = \cos\phi = P / S$

APPARENT POWER

18. $S = V \times I$

REACTIVE POWER

19. $Q = V \times I \times \sin\phi$

SINGLE PHASE POWER

20. $P = V \times I \times \cos\phi$

THREE PHASE POWER

21. $P = \sqrt{3} \times V_L \times I_L \times \cos\phi$

THREE PHASE CURRENT

22. $I_L = P / (\sqrt{3} \times V_L \times \cos\phi)$

TRANSFORMER RATIO

23. $V_1 / V_2 = N_1 / N_2$

TRANSFORMER EFFICIENCY

25. $\eta = \text{Output power} / \text{Input power} \times 100\%$

CAPACITOR

26. $C = Q / V$

27. $X_C = 1 / (2\pi fC)$

28. $U = \frac{1}{2} C V^2$

INDUCTOR

29. $X_L = 2\pi fL$

30. $W = \frac{1}{2} L I^2$

SERIES RESISTANCE

31. $R_{eq} = R_1 + R_2 + R_3$

PARALLEL RESISTANCE

32. $1 / R_{eq} = 1 / R_1 + 1 / R_2 + 1 / R_3$

VOLTAGE DIVIDER

33. $V_{out} = V_{in} \times R_2 / (R_1 + R_2)$

CURRENT DIVIDER

34. $I_1 = I_t \times R_2 / (R_1 + R_2)$

CONDUCTANCE

35. $G = 1 / R$

RESISTIVITY

36. $\rho = R A / L$

CAPACITOR CURRENT

38. $I = C \times dV/dt$

INDUCTOR VOLTAGE 39. $V = L \times dI/dt$

RMS VOLTAGE 40. $V_{rms} = V_{max} / \sqrt{2}$

RMS CURRENT 41. $I_{rms} = I_{max} / \sqrt{2}$

AVERAGE VOLTAGE 42. $V_{avg} = 2V_{max} / \pi$

AVERAGE CURRENT 43. $I_{avg} = 2I_{max} / \pi$

DC MOTOR POWER 44. $P = V \times I$

TORQUE 45. $T = P / \omega$

EFFICIENCY 46. $\eta = \text{Output} / \text{Input}$

BATTERY POWER 47. $P = V \times I$

ELECTRICAL WORK 48. $W = V \times I \times t$

VOLTAGE DROP 49. $V_{drop} = I \times R$

POWER LOSS 50. $P_{loss} = I^2 \times R$